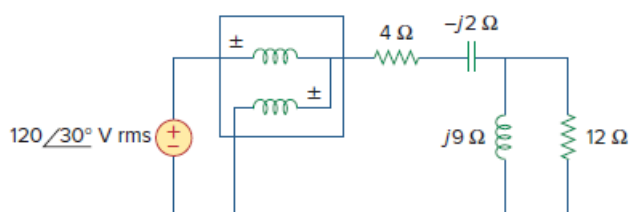


Problem

For the circuit in Fig. 11.33, find the wattmeter reading.

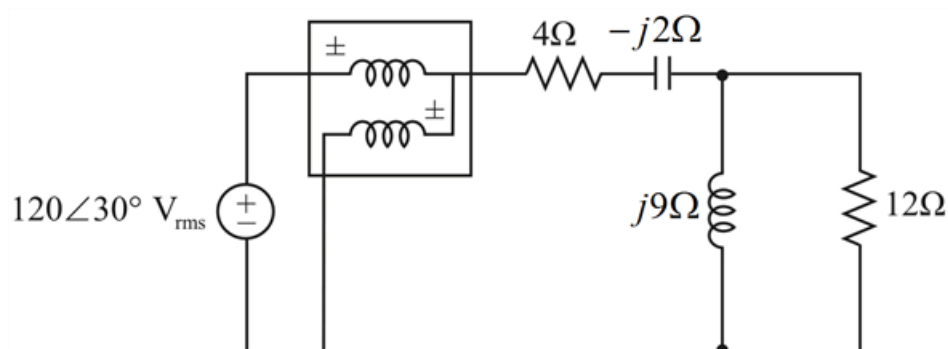
Figure 11.33:



Step-by-step solution

Step 1 of 12

The given circuit is



Step 2 of 12

The Wattmeter reads the average power delivered by the voltage source because the current coil is in series with the voltage source while the voltage coil is in parallel with it.

The 4Ω and $-j2\Omega$ are in series, so their combined impedance Z_1 is

$$Z_1 = 4 + (-j2)$$

$$Z_1 = 4 - j2 \Omega$$

Step 3 of 12

The 12Ω and $j9\Omega$ are in parallel, so their combined impedance Z_2 is

$$Z_2 = \frac{12 \times j9}{12 + j9}$$

$$Z_2 = \frac{j36}{4 + j3} \Omega$$

Z_1 and Z_2 are in series, their equivalent impedance is

$$Z = Z_1 + Z_2$$

Step 4 of 12

$$Z = \frac{j36}{4+j3} + 4 - j2$$

$$Z = \frac{j36+16+j12-j8+6}{4+j3}$$

$$Z = \frac{22+j40}{4+j3} \times \frac{4-j3}{4-j3}$$

$$Z = \frac{208+j94}{25}$$

Step 5 of 12

Thus, the total impedance is

$$Z = (8.32 + j3.76) \Omega$$

The current through the voltage source is,

$$I_{\text{rms}} = \frac{V_{\text{rms}}}{Z}$$

$$I_{\text{rms}} = \frac{120 \angle 30^\circ}{8.32 + j3.76}$$

$$I_{\text{rms}} = \frac{120 \angle 30^\circ}{9.13 \angle 24.32^\circ}$$

$$I_{\text{rms}} = 13.143 \angle 5.68^\circ \text{ A}$$

Step 6 of 12

The complex power is

$$S = V_{\text{rms}} I_{\text{rms}}^*$$

$$S = (120 \angle 30^\circ)(13.143 \angle -5.68^\circ)$$

$$S = 1577.16 \angle 24.32^\circ \text{ VA}$$

$$S = 1437.2 + j649.5 \text{ VA}$$

The wattmeter reads

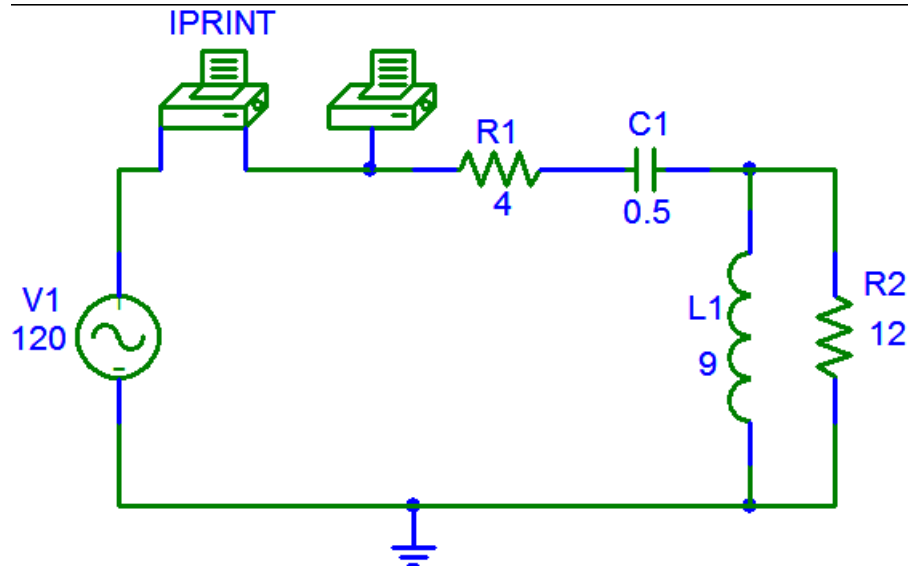
$$P = \text{Re}\{S\}$$

$$\therefore \boxed{P = 1437.2 \text{ W}}$$

Step 7 of 12

We check our results by using PSpice.

The given circuit is drawn using schematics as shown below.



Step 8 of 12

Double click on VAC and enter

$$\text{ACMAG} = 120$$

$$\text{ACPHASE} = 30$$

Consider $\omega = 1 \text{ rad/sec}$, then the inductance and capacitance values will be

$$X_L = \omega L$$

$$9 = (1)L$$

$$\therefore L = 9\text{H}$$

Step 9 of 12

$$X_c = \frac{1}{\omega C}$$

$$2 = \frac{1}{1 \times C}$$

$$\therefore C = 0.5\text{F}$$

Double click on IPRINT and enter

$$\text{AC} = \text{ok}$$

$$\text{MAG} = \text{ok}$$

$$\text{PHASE} = \text{yes}$$

Step 10 of 12

Double click on **VPRINT1** and enter

AC = ok

MAG = ok

PHASE = yes

Once the circuit is drawn and the attributes are set, **Click → Analysis → setup** disable **Bias point Detail** and enable **AC sweep**

Enter Total pt: 1

Start Frequency: 0.159 $\left(\because \frac{1}{2\pi} = 0.159 \text{ Hz} \right)$

End Frequency: 0.159

Step 11 of 12

The schematic is saved, and simulated by selecting **Analysis/Simulate**. The results are saved in output as:

FREQ VM(\$N_0002) VP(\$N_0002)

1.590E-01 1.200E+02 3.000E+01

and

FREQ IM(V_PRINT1) IP(V_PRINT1)

1.590E-01 1.315E+01 5.687E+00

Step 12 of 12

To check the answer, we need the magnitude of the current flowing through the total impedance.

$$I = 13.15 \text{ A}$$

$$P = (I)^2 (R_m)$$

$$P = (13.15)^2 (8.32)$$

$$\therefore \boxed{P = 1438.7 \text{ W}}$$

Hence, satisfied.